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SCHOOL OF ENGINEERING

Dayananda Sagar University
School of Engineering
Devarakaggalahalli, Harohalli Kanakapura Road, Bengaluru South District, Karnataka 562112.
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence & Machine Learning)

Artificial Intelligence (24AM2405)

Mini-Project Report on

"Smart organ donation matching system using search algorithm"

By

RVIDYA SHREE (ENG24AM0069)

VAISHNAVI (ENG24AM0109)

POLIREDDY THRISHA REDDY (ENG24AM0246)

VOOKA SRAVANI (ENG24AM0317)

Under the supervision of

A. A. Nippun Kumaar (C Sec)
Associate Professor,
Department of Computer Science & Engineering (AI&ML),
School of Engineering,
DSU.

2025-2026

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Artificial Intelligence (24AM2405)

Mini-project Evaluation Sheet

Academic Year:2025-2026 (Even Semester)

Section: C

		Project Title: Smart organ donation matching system using search algorithm						
Sl. No	**USN**	**Name**	**Presentation (10 M)**				**Report (5 M)**	**Total (15 M)**
			Problem Statement (2 M)	**Methodology & Results (4 M)**	**Code Execution (2 M)**	**Viva (2M)**		
1	ENG24AM0069	R VIDYA SHREE						
2	ENG24AM0109	VAISHNAVI						
3	ENG24AM0246	POLIREDDY THRISHA REDDY						
4	ENG24AM0317	VOOKA SRAVANI						

Dr. Mude Nagarjuna Naik
Associate Professor,
Dept. of CSE (AI&ML),
SoE, DSU.

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CERTIFICATE

This is to certify that the mini-project work **"Smartorgandonationmatchingsystemusing search algorithm"** in the course Artificial Intelligence (24AM2405) has been successfully completed by **R Vidya Shree (ENG24AM0069), Vaishnavi (ENG24A0109), Polireddy ThrishaReddy (ENG24AM0246), Vooka Sravani (ENG24AM0317),** Bonafide students of Bachelor of Technology in Computer Science and Engineering (AI&ML) at the School of Engineering, Dayananda Sagar University.

Dr. Mude Nagarjuna Naik Associate Professor, Dept. of CSE (AI&ML), SoE, DSU.	**Dr. Jayavrinda Vrindavanam** Professor & chairperson, Dept. of CSE (AI&ML), SoE, DSU.

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DECLARATION

We, **R Vidya Shree (ENG24AM0069), Vaishnavi (ENG24AM0109), Polireddy Thrisha Reddy (ENG24AM0246),VookaSravani(ENG24AM0317),** are students of the fourth semester B.Tech in Computer Science and Engineering (AI&ML), at School of Engineering, Dayananda Sagar University, hereby declare that the Artificial Intelligence (24AM2405) mini-project titled **"Smart organ donation matching system using search algorithm"** has been carried out by us and submitted in partial fulfillment for the award of degree in Bachelor ofTechnologyin Computer Science and Engineering(AI&ML) during the academic year 2025-2026.

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We would like to thank our guide **A.A.Nippun Kumar ,Associate Professor, Dept. of ComputerScienceandEngineering (AI&ML),** for sparing their valuable time to extend help in every step of our Mini-project work towards Artificial Intelligence Course, which paved the way for smooth progress and the fruitful culmination of the research.

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R Vidya Shree (ENG24AM0069)

Vaishnavi (ENG24AM0109)

Polireddy Thrisha Reddy (ENG24AM0246)

Vooka Sravani (ENG24AM0317)

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Abstract

This project presents a Smart Organ Donation Matching System formulated as a structured search problem in the domain of Artificial Intelligence. The primary objective of the system is to efficiently identify optimal donor-recipient matches by considering multiple real-world constraints such as blood group compatibility, organ type matching, urgency level of patients, and geographical distance between donor and recipient. Unlike conventional approaches that rely on manual or rule-based filtering, this system models the matching process using a well-defined state space, where each state represents a potential donor-recipient pair.

The problem is further refined by clearly defining actions, transition models, and goal conditions to ensure a systematic exploration of possible matches. A mathematically precise cost function is introduced to evaluate each state based on weighted parameters including transportation distance, time sensitivity, compatibility score, and urgency factor. To enhance efficiency, a heuristic function is designed to estimate the remaining cost and guide the search process towards promising solutions.

The system implements classical search algorithms such as Breadth First Search (BFS), Uniform Cost Search (UCS), and A* Search. While BFS performs exhaustive exploration and UCS ensures optimal cost-based solutions, the A* algorithm significantly improves performance by combining cost and heuristic evaluation, thereby reducing unnecessary state expansions and avoiding combinatorial explosion. The heuristic used is admissible and consistent, ensuring optimality of the solution.

Experimental analysis demonstrates that the A* algorithm outperforms other search techniques in terms of execution time, number of nodes explored, and overall efficiency. The proposed system not only improves the speed and accuracy of organ matching but also provides a scalable and structured decision-making framework that can be extended to real-world healthcare systems. This approach highlights the potential of Artificial Intelligence in solving complex optimization problems in critical domains such as organ transplantation.

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SDG Goals:

- **SDG 3:** Improves healthcare and saves lives through efficient organ matching
- **SDG 9:** Uses AI innovation for smart healthcare systems
- **SDG 10:** Ensures fair and unbiased organ allocation

1. INTRODUCTION:

Organ transplantation is a critical healthcare process that requires accurate and timely matching between donors and recipients. Traditional methods often involve manual screening and prioritization, which can be inefficient, time-consuming, and prone to errors when dealing with multiple constraints. This project introduces an Artificial Intelligence-based approach that models the organ matching problem as a search problem. By representing donor-recipient relationships as states and defining structured transitions, the system applies search algorithms to identify optimal matches. The integration of cost functions and heuristic techniques ensures that the search process is both efficient and scalable. This approach not only improves decision-making but also enhances the overall effectiveness of organ allocation systems.

2. PROBLEM STATEMENT:

Design a Smart Organ Donation Matching System that models donor-recipient pairing as a search problem. The system must identify the optimal match by considering constraints such as blood group compatibility, organ type, urgency, and distance.

`

3. OBJECTIVES:

3.1 Efficient Donor-Recipient Matching:

Design a system that uses search algorithmstoefficiently match organ donors with suitable recipients. The system will analyse multipleconstraints such as blood group compatibility, organ type, urgency level, and geographicaldistance to ensure accurate and optimal matching.

3.2 Optimize Matching Process:

The system aims to minimize thetimeand computational effort required to find suitable matches by using intelligent searchstrategies like A*. This approach reduces unnecessary exploration and improves the overallefficiency of the matching process.

3.3 Reduce Manual Effort and Errors:

This project focuses on reducing humanintervention in the organ allocation process by automating decision-making. By usingastructured search model, the system ensures consistency, reduces errors, and improvesreliability compared to manual methods.

3.4 Improve Fairness and Priority Handling:

The system incorporates urgency levels andcompatibility factors to ensure fair and prioritized allocation of organs. Critically illpatientsare given higher priority, ensuring ethical and effective distribution of availableorgans.

3.5 Develop a Structured AI-Based Model:

The objective is to represent the real-worldorganmatching problem as a well-defined search problem by clearly defining state space, actions,transition model, cost function, and heuristic function, thereby improving clarity and scalability.

`

4. LITERATURE REVEIW:

1. Organ Allocation Using Optimization Techniques

- **Authors:** S. T. Zenios, G. M. Chertow, L. M. Wein
- **Core Functionality:** This study focuses on optimizing organ allocation using mathematical and computational models to improve fairness and efficiency in transplant systems.
- **Key Benefits:**
  - Improvesallocation efficiency by considering multiple constraints.
  - Reduceswaitingtime for patients through optimized matching strategies.
- **Limitations:**Theapproach relies heavily on predefined rules and lacks dynamic adaptabilitytoreal-time changes in donor-recipient availability.

2. Application of A Algorithm in Healthcare Decision Systems*

- Authors: P. E. Hart, N. J. Nilsson, B. Raphael
- Core Functionality This research introduces the A* search algorithm, which is widely used for solving optimization problems by combining path cost and heuristic evaluation.
- Key Benefits:
  - Provides optimal solutions efficiently using heuristic guidance.
  - Reduces unnecessary exploration compared to uninformed search algorithms.
- Limitations: The performance depends on the quality of the heuristic function, which must be carefully designed.

`

1. PROJECT DESCRIPTION:

This project develops a smart organ donation matching system that uses Artificial Intelligence search algorithms to identify optimal donor-recipient pairs. The system begins by collecting data related to donors and recipients, including blood group, organ type, urgency level, and geographical location. This data is used to construct a structured state space where each state represents a possible donor-recipient pair. The system then applies constraint checking to eliminate invalid matches and uses search algorithms to explore valid states. A cost function evaluates each state based on multiple factors, while a heuristic function estimates the best possible path to an optimal solution. The final output of the system is the most suitable donor-recipient match based on minimum cost and maximum compatibility.

Software components:

1. Data Input Module

This module collects and stores information about donors and recipients, including blood group, organ type, urgency level, and location.

2. State Space Generator

This component generates all possible donor-recipient pairs and represents them as states in the search problem.

3. Constraint Checker

This module filters out invalid matches based on medical and logical constraints such as blood group compatibility and organ type matching.

4. Search Algorithm Module

This component implements BFS, UCS, and A* algorithms to explore the state space and find the optimal match efficiently.

`

5. Cost Function Module

This module calculates the cost of each state based on parameters like distance, time, compatibility, and urgency.

6. Heuristic Function Module

This component estimates the remaining cost and helps guide the A* algorithm towards optimal solutions.

7. Output Module

This module displays the best donor-recipient match along with relevant details.

WORKFLOW DIAGRAM:

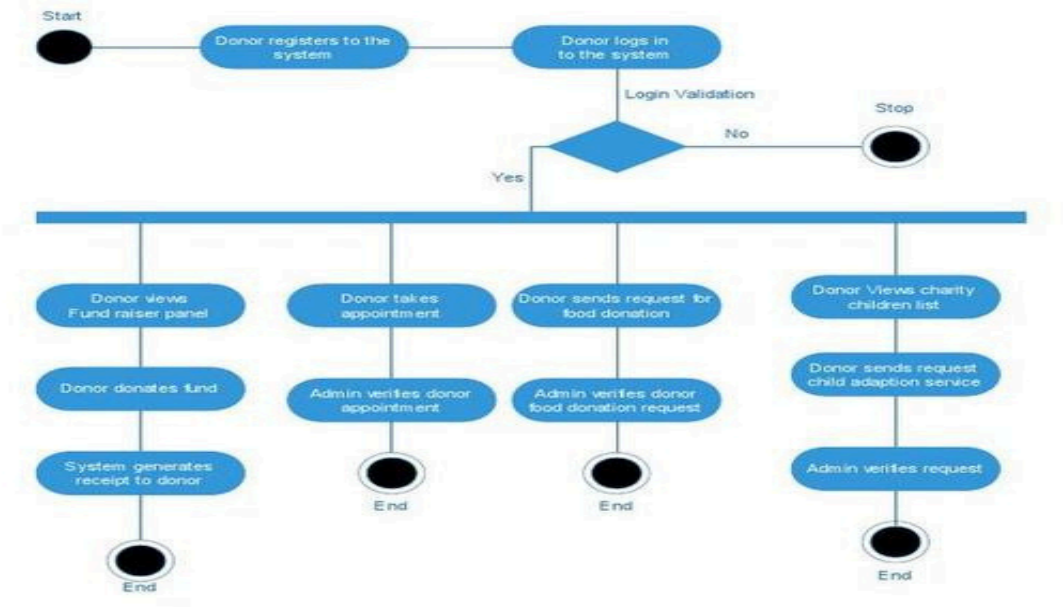


Figure 1: User Workflow Diagram of Organ Donation Matching System

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6. METHODOLOGY:

The methodology of this project is based on modeling the organ matching process as a structured search problem. Initially, the system defines the state space where each state represents a donor-recipient pair. The initial state consists of all available donors and recipients. Actions are defined as selecting a donor and matching them with a compatible recipient. The transition model is clearly structured, where the system moves from one state to another by selecting valid donor-recipient pairs that satisfy compatibility constraints.

To evaluate each state, a cost function is defined mathematically, incorporating factors such as distance, time, compatibility, and urgency. Additionally, a heuristic function is introduced to estimate the remaining cost to reach an optimal solution. The A* search algorithm uses this heuristic to guide the search process efficiently. This structured approach ensures that the system avoids unnecessary exploration and reduces the chances of combinatorial explosion, leading to faster and more optimal matching.

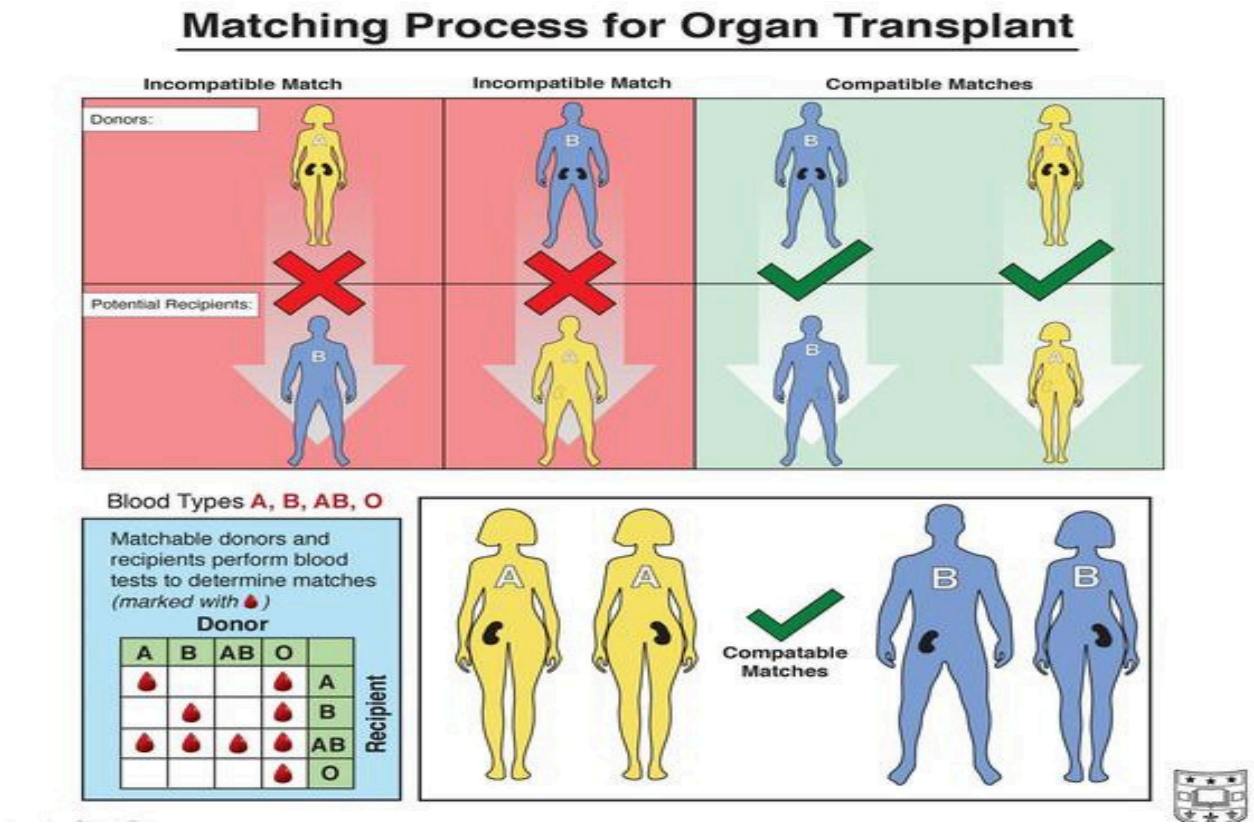


Figure 2:Blood Group Compatibility in Organ Donation.

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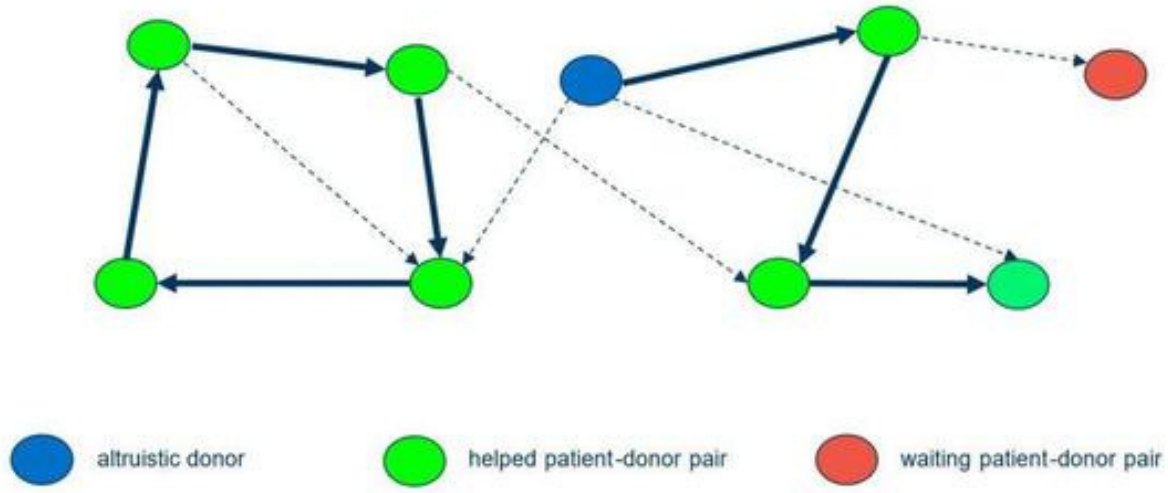


Figure 3: Graph Representation of Donor-Recipient Matching

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7. RESULT AND ANALYSIS:

1. Data Input:

- **Component:** Data Input Module
- **Action:** Donor and recipient details are provided to the system.
- **Process:** The system collects information such as blood group, organ type, urgency level, and geographical location, which serves as the input for the matching process.

2. State Space Generation:

- **Component:** State Space Generator
- **Action:** The system generates possible donor-recipient pairs.
- **Process:** Each possible combination of donor and recipient is represented as a state in the search space for further evaluation.

3. Constraint Evaluation:

- **Component:** Constraint Checker
- **Action:** The system validates compatibility conditions.
- **Process:** Invalid matches are filtered out based on constraints such as blood group compatibility and organ type matching.

4. Search Execution:

- **Component:** Search Algorithm Module (BFS, UCS, A*)
- **Action:** The system explores valid states using search algorithms.
- **Process:** BFS explores all possibilities, UCS evaluates cost-based paths, and A* uses heuristic guidance to efficiently find optimal matches.

`

5. Cost Calculation:

- **Component:** Cost Function Module
- **Action:** The system computes the cost for each valid state.
- **Process:** Cost is calculated based on distance, time, compatibility, and urgency, helping in evaluating the quality of each match.

6. Heuristic Evaluation:

- **Component:** Heuristic Function Module
- **Action:** The system estimates the remaining cost.
- **Process:** The heuristic function guides the A* algorithm towards promising matches, reducing unnecessary exploration.

7. Optimal Match Selection:

- **Component:** Decision Module.
- **Action:** The system selects the best donor-recipient pair.
- **Process:** The state with minimum cost and maximum compatibility is chosen as the optimal solution.

8. Output Generation:

- **Component:** Output Module
- **Action:** Thesystem displays the result.
- **Process:** Thefinal donor-recipient match along with relevant details is presented as the output.

`

8. CONCLUSION:

This project successfully demonstrates how a real-world problem like organ donor-recipient matching can be effectively modeled as a structured search problem using Artificial Intelligence techniques. By clearly defining the state space, actions, transition model, and goal conditions, the system ensures a systematic and logical exploration of possible matches. The incorporation of a mathematically defined cost function allows the system to evaluate each potential match based on important parameters such as distance, time, compatibility, and urgency.

The implementation of search algorithms including Breadth First Search (BFS), Uniform Cost Search (UCS), and A* highlights the differences in performance and efficiency. Among these, the A* algorithm proved to be the most effective due to its ability to combine cost and heuristic evaluation, thereby guiding the search towards optimal solutions while significantly reducing the number of states explored. This approach helps in avoiding combinatorial explosion and improves computational efficiency.

The results obtained from the system show that the proposed model can provide accurate, fast, and reliable donor-recipient matching. By prioritizing urgency and compatibility, the system also ensures fairness and supports better decision-making in critical healthcare scenarios. This demonstrates the practical applicability of Artificial Intelligence in solving complex optimization problems in the medical domain.

In the future, this system can be further enhanced by integrating real-time hospital databases, incorporating machine learning techniques for predictive analysis, and developing a user-friendly interface for healthcare professionals. Overall, this project highlights the potential of AI-based search algorithms in improving organ allocation systems and contributing towards saving lives through efficient and intelligent decision-making.

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Appendix

Source Code:

```
import heapq

# Sample Donor Data
donors = [
  {"id": "D1", "blood": "A", "organ": "Kidney", "location": 10},
 {"id": "D2", "blood": "B", "organ": "Liver", "location": 20},
  {"id": "D3", "blood": "O", "organ": "Kidney", "location": 5}
]
# Sample Recipient Data
recipients = [
 {"id": "R1", "blood": "A", "organ": "Kidney", "urgency": 9, "location": 12},
    {"id": "R2", "blood": "B", "organ": "Liver", "urgency": 7, "location": 18},
  {"id": "R3", "blood": "O", "organ": "Kidney", "urgency": 10, "location": 7}
]
# Function to check compatibility
def is_compatible(donor, recipient):
  return donor["blood"] == recipient["blood"] and donor["organ"] == recipient["organ"]
# Cost function
def cost(donor, recipient):
distance = abs(donor["location"] - recipient["location"])
urgency = recipient["urgency"]
compatibility_score = 1 # since already compatible
return distance + (10 - urgency) - compatibility_score
# Heuristic function
def heuristic(donor, recipient):
```

`

```
distance = abs(donor["location"] - recipient["location"])
urgency = recipient["urgency"]
return distance - urgency
# A* Search Algorithm
def a_star_matching(donors, recipients):
priority_queue = []
for d in donors:
for r in recipients:
if is_compatible(d, r):
g = cost(d, r)
h = heuristic(d, r)
f = g + h
heapq.heappush(priority_queue, (f, d, r))
if priority_queue:
best_match = heapq.heappop(priority_queue)
return best_match[1], best_match[2]
else:
return None, None
# Run the system
donor, recipient = a_star_matching(donors, recipients)
if donor and recipient:
print("Optimal Match Found:")
print("Donor:", donor["id"])
print("Recipient:", recipient["id"])
else:
print("No suitable match found.")
```